

How the system *reads* the market.

A reference for the signal pipeline — the data, the indicators, the regime engine, and the discipline that holds them together.

MARKET
REGIMES

7

Read daily

TRAILING
STOP

30%

Hard exit

SYMBOLS

~4,000

Scanned daily

WALK-
FORWARD

21y

Continuous,
no cherry-
picking

What this document *covers*.

A technical reference for the RigaCap signal engine: how raw market data is acquired, how indicators are computed, how the entry ensemble decides, how regimes shape behavior, and how positions are managed once they are live.

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One sentence on posture. Every figure in this document is drawn from walk-forward simulation across multiple start dates — the average, not a cherry-picked best case. Production parameters are locked. The walk-forward harness has additional research-mode capabilities, and where this document mentions them they are flagged as such.

The data *pipeline*.

The pipeline begins with a universe of roughly four thousand US equities. Every trading afternoon the system fetches, validates, and enriches the price record before it ever reaches the signal engine.

Stock universe.

PARAMETER	VALUE	SOURCE
Total universe	~4,000 symbols	NASDAQ + NYSE
Minimum price	\$15	<code>config.MIN_PRICE</code>
Minimum volume	500,000 shares/day	<code>config.MIN_VOLUME</code>
Lookback	7 years OHLCV	Pickle on S3, Parquet shadow
Primary source	Alpaca SIP	Consolidated, split-adjusted
Fallback	yfinance	Indexes (^VIX, ^GSPC), failover

Universe hygiene.

All exchange-traded funds are excluded from the signal universe. The exclusion is categorical — leveraged products, sector ETFs, broad-index ETFs, crypto-linked products, and thematic baskets are all out. Ordinary operating companies, including miners, are eligible. The exclusion is enforced at the symbol-list level and reasserted at scan time so a recategorised ticker cannot leak into the universe between rebuilds.

Fetch architecture.

PARAMETER	VALUE	PURPOSE
Batch size	100 symbols (Alpaca) / 10 (yfinance)	Provider-appropriate batching
Max retries	2 per batch	Transient failure handling
Batch delay	1.5 s	Rate-limit compliance
Retry delay	3.0 s (2x batch delay)	Backoff on failure
Required symbols	SPY, ^VIX	Always fetched (regime detection)

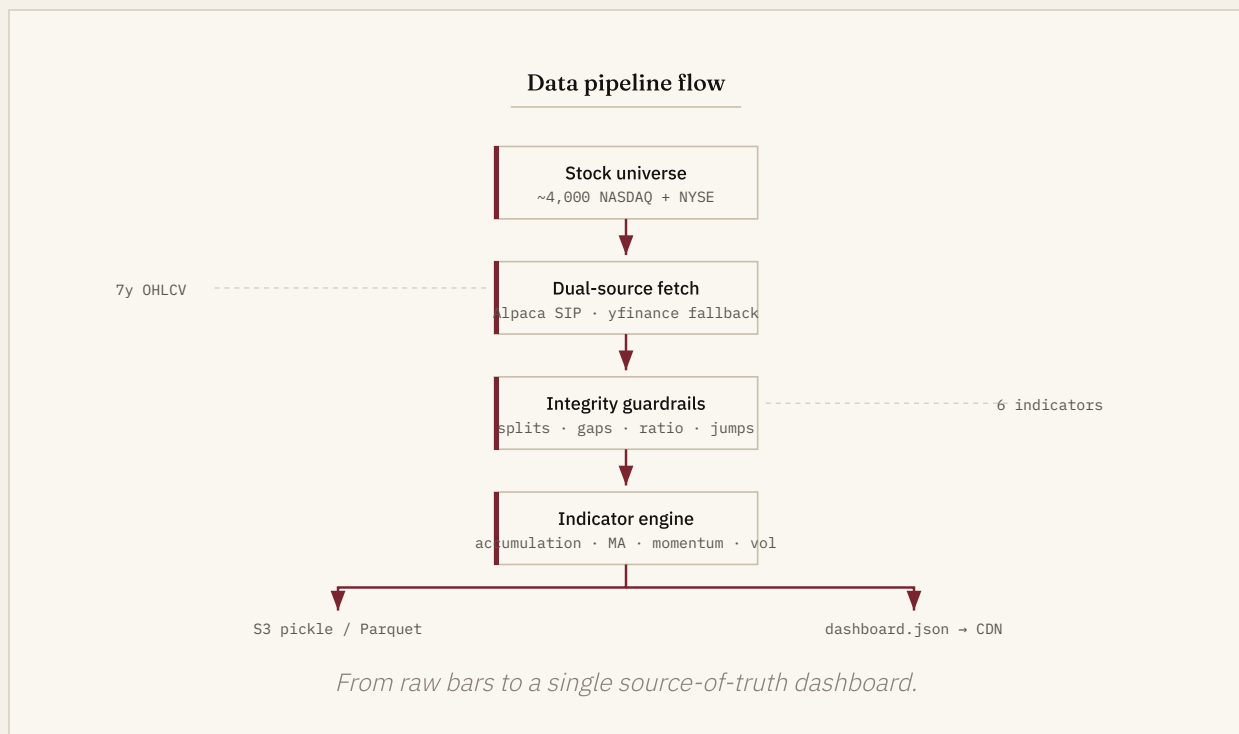
After the initial multi-year fetch, daily updates run in **incremental mode** — only prices since the last cached date are downloaded. Updates that took minutes on a cold load take seconds incrementally. The full historical record is persisted to S3 as a consolidated gzipped pickle (durable cache for Lambda cold-starts) with a parallel Parquet shadow write that supports ad-hoc SQL via DuckDB.

Data-integrity guardrails.

Split-adjustment and corruption detection are now built into every layer of the pipeline. End-to-end split-adjustment eliminates a class of historical-data artifacts that previously could distort returns or trigger spurious exits.

- **Split-adjusted by default.** The Alpaca `StockBarsRequest` uses `Adjustment.SPLIT` on every call, so a 10-for-1 or 20-for-1 split cannot create an artifact -90% drop in the historical series.
- **Abrupt-jump rejection.** Any single-day close-to-close change greater than 65% (log-return > 0.5) marks the symbol as corrupted. Real market moves rarely exceed this; unadjusted corporate actions always do.

- **Ratio sanity.** Symbols where the current close is more than 50x or less than 0.02x the long-term accumulation reference are quarantined as clearly corrupted — typically ticker reuse or a missed adjustment.
- **Long-gap rejection.** Any symbol with a gap longer than ten trading days between consecutive bars is flagged as ticker-reuse corruption (old entity delisted, ticker recycled to a new entity).
- **Nightly Layer 2 hygiene.** Alpaca's corporate-actions API is polled every weekday at 6:30 PM ET. Splits trigger automatic force-refetch with split adjustment. Ticker reuses trigger auto-quarantine. An admin digest reports outcomes.
- **Asset-ID tracking.** Every symbol's Alpaca UUID is stored in `symbol_metadata` . When an exchange recycles a ticker, the UUID changes and the system detects the swap before the next signal fires.



The indicator *engine*.

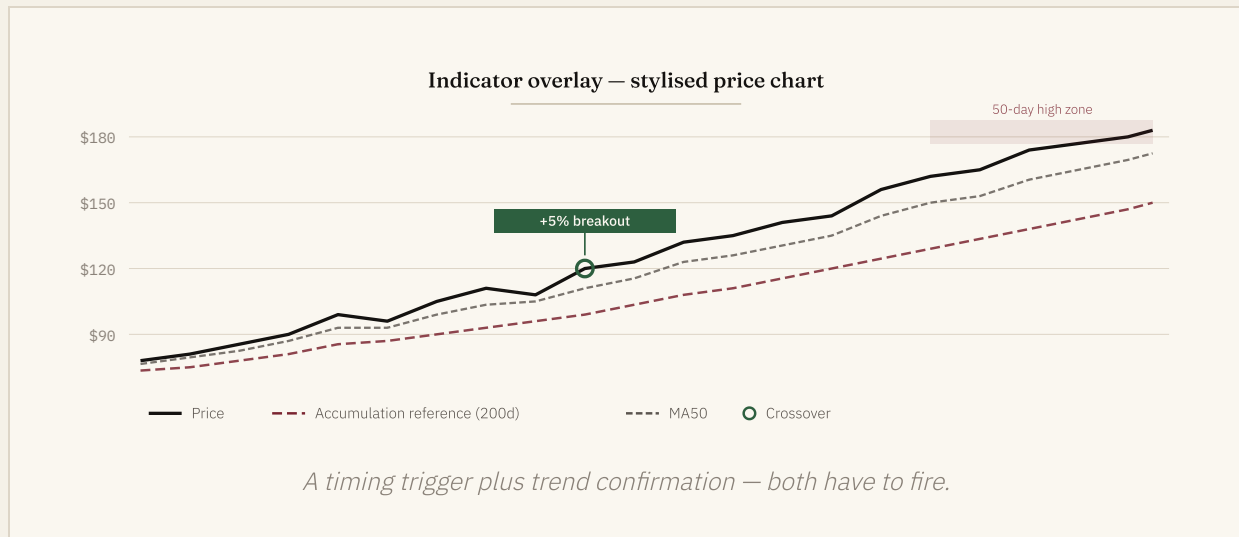
A short list of indicators — computed once per scan, cached on demand — feeds both the timing layer and the momentum-quality ranking.

INDICATOR	FORM	PERIOD	ROLE
Accumulation reference	$\frac{\sum(\text{price} \times \text{volume})}{\sum(\text{volume})}$	200d	Volume-weighted timing baseline
Moving averages	Simple	20 / 50 / 200d	Trend confirmation, quality filter
Short momentum	$(p / p[t-5] - 1)$	5d	Short-term acceleration · weight 0.3
Long momentum	$(p / p[t-60] - 1)$	60d	Sustained trend strength · weight 0.2
Volatility	$\sigma(\text{returns}, 20) \times \sqrt{252}$	20d	Risk penalty in composite · weight 0.15
Distance from high	$p / \max(p, 50d) - 1$	50d	Breakout proximity

Lazy computation. Indicators are computed on demand and cached in the per-scan data structure. The first scan pays for the indicator pass; subsequent

operations within the same scan reuse cached values.

The accumulation reference is RigaCap's proprietary timing baseline — a long-term volume-weighted price that filters out noise from short-term spikes. The breakout filter fires when price clears that baseline by a small, conviction-weighted margin and the rest of the indicator stack confirms.



The ensemble — *three* factors.

A candidate must satisfy three independent factors before the system generates an entry. Multi-confirmation reduces false signals and keeps the bar high.

Factor 1 — timing.

A proprietary breakout detector fires when price clears the long-term accumulation reference by a conviction-weighted margin. The threshold is a fixed lever in production (the 5% level), tuned during walk-forward research and locked at deployment. It catches early breakouts before momentum-only systems acknowledge them.

Factor 2 — momentum quality.

Candidates that pass timing are ranked by a composite momentum score combining short-term acceleration (5-day), longer-term trend strength (60-day), and a volatility penalty. The production formula, exactly as it runs live:

$$\text{score} = 0.3 \times r_{5d} + 0.2 \times r_{60d} - 0.15 \times \sigma_{20d}$$

where r_{5d} and r_{60d} are trailing returns in percent and σ_{20d} is 20-day annualised volatility in percent. Only top-ranked names are eligible. Note the weighting: the **short horizon carries the higher weight** — the composite rewards fresh acceleration within an established 60-day trend, while the volatility penalty disqualifies candidates whose “momentum” is really just whipsaw. Candidates must additionally trade above their 20-day and 50-day moving averages (the trend-quality gate).

Factor 3 — confirmation.

FILTER	CONDITION	PURPOSE
Breakout proximity	Within 3% of 50-day high	High-conviction entries only
Volume confirmation	Day's volume $\geq$ 1.3x recent average	Real participation
Trend	Price > MA20 and MA50	Established uptrend
Liquidity	$\geq$ 500,000 shares/day	Tradable size
Floor	Price $\geq$ \$15	Filters penny stocks

Anti-squeeze filters.

A fourth filter family rejects parabolic candidates that have already run before our other factors fire — a guard against entering at squeeze tops. The filter was added after a 2021 review of the strategy's worst entries.

FILTER	LEVER	PURPOSE
Recent-return cap	<code>max_recent_return_pct</code>	Reject candidates already up beyond a 30-day threshold
Velocity cap	<code>price_velocity_cap_pct</code>	Reject candidates with an extreme 5-day move
Overbought guard	<code>rsi_oversold_filter</code>	Standard RSI ceiling on the entry candidate

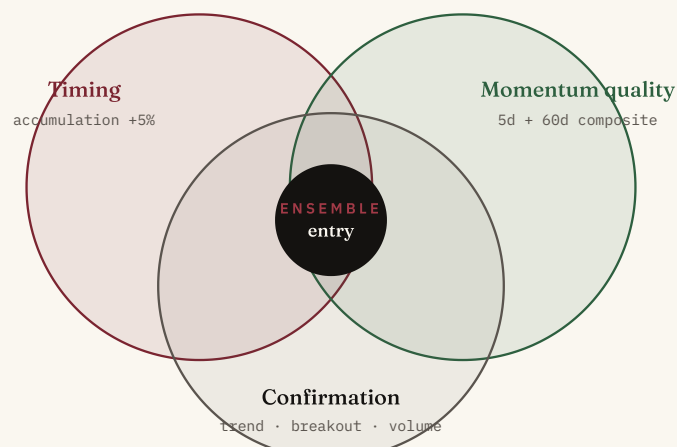
Production parameters.

Production trades the locked RigaCap Momentum Strategy parameter set (live since June 10, 2026): up to twenty positions sized at roughly 4.5% each with inverse-volatility weighting, a 30% trailing stop, and the timing-and-momentum ensemble

described above, drawn from a universe of the top 100 symbols by 60-day volume. The walk-forward harness explores wider ranges in research mode; production does not retune itself between runs.

Research vs. production. The walk-forward harness can re-optimize parameters on a rolling window using Bayesian methods — research capability that informs which parameter set gets locked. Production then runs the locked parameters until a new validation pass justifies replacement. The system does not silently retune itself in front of subscribers.

Three-factor convergence



All three sets must overlap for an entry to fire.

Market regime *intelligence*.

The regime classifier reads SPY price action, VIX levels, market breadth, and new-highs data, and assigns the current environment to one of seven distinct states. The output drives a single, deliberate response in production.

Seven regimes.

Strong Bull SPY clearly above 200-day; VIX low; broad participation.	Weak Bull SPY above 200-day but only modestly; volatility moderate.
Rotating Bull SPY choppy with positive 1-year trend; sector rotation in play.	Range-Bound SPY within a few percent of its 200-day; flat trailing return.
Weak Bear SPY below 200-day, elevated VIX, deteriorating breadth.	Panic / Crash SPY meaningfully below 200-day; VIX in the upper quantile.
Recovery Bouncing from lows; SPY above 50-day but still recovering.	Forecast A 1–2 week transition outlook layered on the static state.

Scoring weights.

Regime detection uses weighted scoring across condition indicators. Higher weights mean that indicator carries more influence:

INDICATOR	WEIGHT	ROLE
Long-term trend (252d SPY return)	2.5x	Differentiator: rotating-bull vs. range-bound
SPY vs. 200MA	1.5x	Primary bull/bear boundary
New highs (%)	1.5x	Breadth confirmation of trend
Breadth (% above 50MA)	1.2x	Width of market participation
VIX percentile, trend strength, SPY vs. 50MA	1.0x	Standard conditions

Regime response.

Production response is deliberately binary: when the regime engine reads SPY below its 200-day moving average, the system moves to cash; otherwise it stays invested at the locked production parameter set. Cascade Guard runs alongside (next section). Per-regime sizing — varying trailing-stop width, position count, or position size by regime state — is a research-mode capability of the walk-forward harness, not a live rule.

REGIME	PRODUCTION ACTION	NOTES
Strong Bull	Invested	Locked parameters
Weak Bull	Invested	Locked parameters
Rotating Bull	Invested	Locked parameters
Range-Bound	Invested unless SPY < 200MA	Locked parameters
Weak Bear	Cash if SPY < 200MA	Capital preservation
Panic / Crash	Cash	SPY below 200-day
Recovery	Invested	Re-engagement on 50MA reclaim

Walk-forward optimizer choices in research mode are sometimes shown per regime. Production currently uses a binary cash-or-invested response with Cascade Guard pause; per-regime sizing is a research-mode capability, not a live rule.

Regime spectrum — production response

Strong Bull	Weak Bull	Rotating	Range	Weak Bear	Panic	Recovery
INVESTED	INVESTED	INVESTED	conditional	SPY < 200MA	CASH	re-engage
LOWER RISK				HIGHER RISK		

A binary cash/invested response, anchored on SPY vs. 200-day.

Transition forecasting.

Beyond static classification, the regime engine computes indicator trajectories to predict where the market is heading in the next one-to-two weeks:

- **VIX delta (5-day).** Rising VIX boosts bearish regime probabilities.
- **Trend acceleration.** Recent 5-day return compared to the prior 5-day return.
- **200MA proximity.** Within 2% of the 200-day raises transition uncertainty.
- **Sharp moves.** 5-day returns exceeding plus-or-minus 5% boost crash and recovery probabilities.

The forecast collapses to a recommended action: **stay_invested** (SPY > 200MA) or **go_to_cash** (SPY < 200MA). No intermediate states.

Trailing stops & *exits*.

The trailing stop is calculated from the high water mark — the highest price since entry, not the entry price itself. Stops ratchet up as a position appreciates and never walk back down.

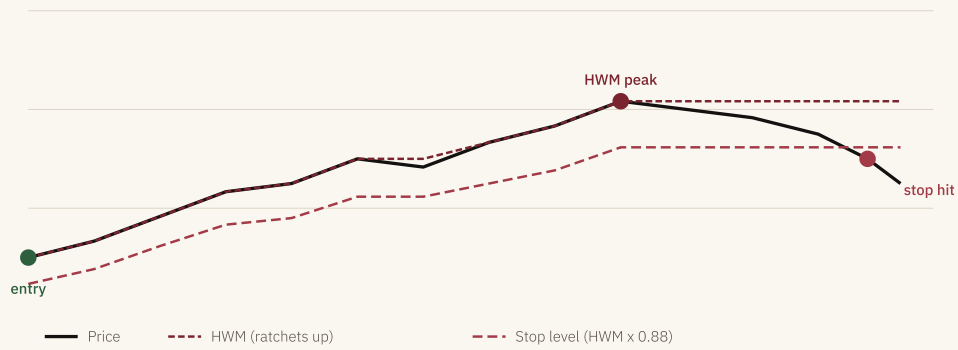
```
# High water mark and trailing stop level
HWM      = max(entry_price, stored_highest, current_price)
stop_level = HWM × (1 - 0.30)

# Exit and warning conditions
SELL      when price ≤ stop_level
WARN      when (price - stop_level) / price < 0.03
```

Exit priority.

1. Regime exit (go_to_cash) — immediate sell regardless of P&L; capital preservation overrides everything else.
2. Trailing-stop hit — price at or below HWM x 0.70; closes the position.
3. Warning zone — within 3% of the stop level; triggers a subscriber alert, no automatic action.
4. Regime warning — SPY approaching its 200-day average; advisory only.
5. Rebalance rotation — at the periodic universe re-ranking, an existing position has dropped out of the top tier and a stronger candidate displaces it. The boundary itself is not an exit trigger; the rank change is.

Trailing stop — high water mark staircase



Stops follow the price up — never down.

Walk-forward to production — exact *parity*.

The walk-forward result is only realizable if production runs the same strategy the backtest validated. Every mechanic — ranking universe, HWM tracking, stop trigger, position sizing — is audited for parity, and any divergence is treated as a defect that costs subscribers real returns.

Most of these alignments were in place at launch. A handful drifted as features were added incrementally and were realigned in mid-May 2026 after this audit surfaced the gaps. The table below is the post-audit state.

Parity table.

MECHANIC	WALK-FORWARD DEFAULT	PRODUCTION LIVE
Ranking universe	top 500 by 60-day avg volume, recomputed per period	top 500 by liquidity (matched May 17 2026 — previously 100)
HWM source	day's close	day's close (from price cache)
HWM persistence	in-process across periods	database commit on every pass (unconditional)
Stop trigger	$\text{close} \leq \text{HWM} \times (1 - \text{stop \%})$	$\text{close} \leq \text{stop level}$ (intraday firing intentionally disabled)
Trailing stop %	strategy parameter (30% in production)	strategy parameter (30% in production)
Max concurrent positions	20	20

MECHANIC	WALK-FORWARD DEFAULT	PRODUCTION LIVE
Position sizing	~4.5% of capital per slot, inverse-volatility weighted	~4.5% of capital per slot, inverse-volatility weighted
Entry: timing reference	price above long-term accumulation reference + breakout band	identical filter
Entry: breakout band	within 3% of 50-day high	identical
Entry: momentum quality	top ensemble score	identical
Carry positions across rebalance	true (positions survive period boundaries)	true (live portfolio is always-on)
Regime adjustments	per-regime deltas to trailing stop & sizing	identical, via dashboard regime cache

The audit method.

Parity is not assumed; it is measured. The largest single realization gap discovered in this audit was the ranking universe: production filtered to 100 most-liquid symbols, while the walk-forward result was generated against 500. A liquidity-rank audit of a representative walk-forward run found that **about 70% of the simulated trades came from symbols ranked 101-500** — including several of the strongest winners.

Production at N=100 was quietly missing the majority of validated trade opportunities. The fix was a one-line configuration change; the discovery required reading the trade list against the same ranking logic and counting where each entry came from.

This kind of mechanical audit runs whenever a new lever is added to either side — backtester or production — before the lever is allowed to influence published results.

Cascade Guard — the portfolio *circuit breaker*.

Most position-level systems can be undone by a single bad day — multiple stops fire, new positions enter into the same fall, and the cycle repeats. Cascade Guard exists to interrupt that pattern.

What it does.

When several positions hit their trailing stop on the same day, Cascade Guard pauses all new entries for ten trading days. The threshold and pause length are conservative, and the mechanism is deliberately rare — it fires on cascade events, not on ordinary drawdowns.

Live in production since May 2026. Cascade Guard has been a fixture of the walk-forward backtest for some time; it now runs in production with state persisted to S3 (`s3://rigacap-prod-price-data-149218244179/cb-state/<portfolio_type>.json`). The change closed the final walk-forward-to-production parity gap.

Mechanics.

LEVER	SETTING	BEHAVIOUR
Trigger threshold	3 same-day trailing-stop hits	Cascade detected

LEVER	SETTING	BEHAVIOUR
Pause length	10 trading days	No new entries during pause
Existing positions	Unaffected	Continue to be monitored normally
HWM & regime checks	Continue to run	Stops still fire if hit
Re-engagement	Automatic	Pause expires; ensemble resumes

What it adds.

Cascade Guard is validated by counterfactual: the walk-forward backtest is re-run with the safeguard disabled and the results compared. A prior strategy configuration showed a meaningful return-quality contribution; the attribution for the current production configuration is being re-measured on the 21-year window before we publish a number. The drawdown impact is essentially neutral, so we frame Cascade Guard as a re-entry discipline, not as drawdown protection.

Cascade Guard fires on rare cascade events — a handful of times across a market cycle. The point is not its frequency; it is the asymmetry. The cost during quiet periods is zero. The benefit during a cascade is avoiding the second wave of re-entries into the same falling market.

Closes the parity gap

Cascade Guard had been a backtest-only mechanism through April 2026. Production now runs the same logic so subscribers experience the lever the simulation measured.

Conservative by design

The trigger and pause length are deliberately not aggressive. The mechanism's job is to interrupt cascades, not to frequently override the strategy.

Intraday *monitoring*.

Once positions are open, the system checks live prices every five minutes during market hours. The job is HWM tracking and regime-change detection — not minute-by-minute trading.

Schedule.

PARAMETER	VALUE
Hours	Monday–Friday, 09:00–15:55 ET
Interval	Every 5 minutes
Price source	Alpaca latest quote (SIP), yfinance fallback
Alert deduplication	<code>(position_id, action, date)</code>

Trailing-stop exits run end-of-day.

After research showed intraday cadence cost roughly 17 percentage points annualised versus end-of-day evaluation, trailing-stop exits are evaluated end-of-day. HWM tracking and regime-exit checks continue to run every five minutes during market hours; the highest price seen each session is captured for the next day's stop calculation, and a regime transition can still close the book intraday. The 5-minute monitor is a state-tracking surface, not an execution loop. The behavior is controlled by a single environment flag (`INTRADAY_TRAILING_STOP_ENABLED=false` in production).

Why end-of-day stops. Intraday execution is exposed to data anomalies and brief flash moves that resolve before the close. Evaluating the trailing stop on the daily close removes that noise without sacrificing the discipline — the stop still fires the same day, just on a confirmed price.

Watchlist intraday signal detection.

The monitor also watches the dashboard's watchlist — candidates approaching the timing threshold but not yet through it. If a watchlist candidate clears the threshold intraday, the system promotes it to a fresh signal, removes it from the watchlist, alerts subscribers who do not already hold it, and updates the S3 dashboard JSON in place.

Position *management.*

The model portfolio service maintains parallel ledgers — a live portfolio that mirrors production behavior, a walk-forward portfolio for performance measurement, and three ghost portfolios that document configuration alternatives.

Production parameters.

20 MAX POSITIONS	4.5% POSITION SIZE (INVERSE-VOL)	30% TRAILING STOP	Top 100 UNIVERSE BY 60- DAY VOLUME
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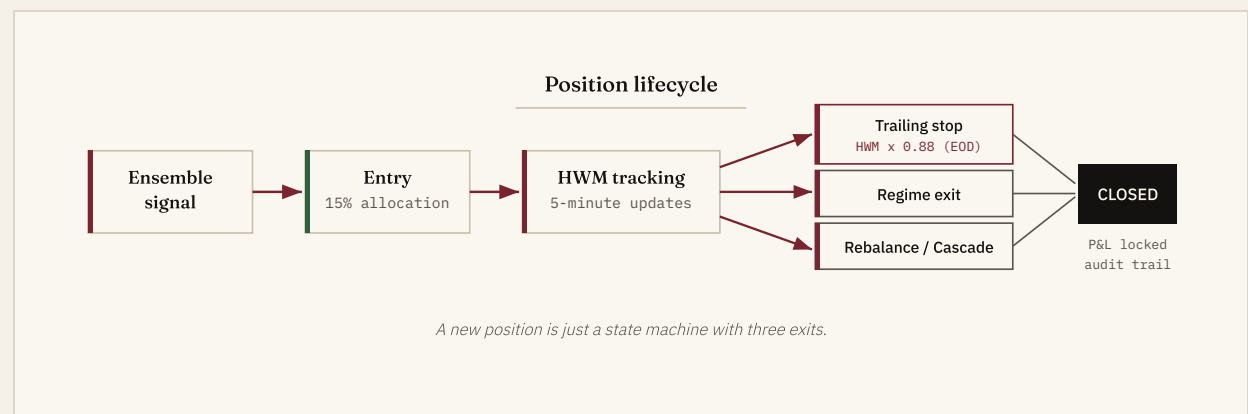
Entry logic.

- Read fresh ensemble signals from the dashboard cache (S3 JSON).
- Filter to `is_fresh = true` only; skip symbols already held.
- Sort by ensemble score, highest first.
- Enter up to `MAX_POSITIONS - open_count` new positions.
- Allocate roughly 4.5% per position, weighted by inverse volatility so calmer names carry more.
- Minimum allocation \$100 to avoid sub-share rounding.

Exit paths.

EXIT REASON	TRIGGER	PORTFOLIO
<code>trailing_stop</code>	Daily close $\leq$ HWM $\times$ 0.70	Live + WF
<code>regime_exit</code>	<code>go_to_cash</code> recommendation	Live
<code>rebalance_exit</code>	Period boundary (walk-forward harness)	Walk-forward
<code>cascade_pause</code>	Cascade Guard active during entry window	Live + WF

Re-ranking boundaries. “Rebalance” here means a universe re-ranking, not a portfolio shuffle for the subscriber. The system periodically re-runs the entry gauntlet across the full universe, identifies the current top-tier candidates, and checks whether existing held positions still qualify. Positions that still rank carry forward; ones that have dropped out of the top tier rotate out. Carry-on semantics outperform force-close, so the system carries qualifying positions across windows rather than mechanically liquidating at the boundary.



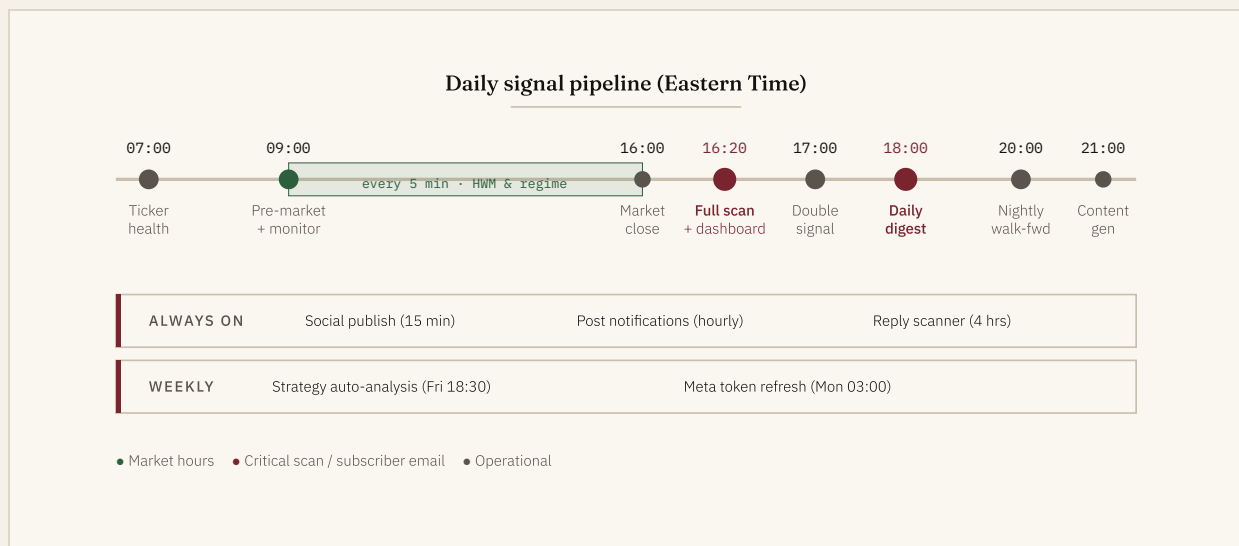
Ghost portfolios.

Three alternative parameter sets run alongside the production portfolio for ongoing comparison — the goal is to document what was rejected and why:

GHOST	TRAILING STOP	MAX POSITIONS	POSITION SIZE	STYLE
Aggressive	8%	8	12%	Tight stops, more positions
Conservative	18%	4	20%	Wide stops, fewer positions
Top-3 only	12%	3	30%	Highest-conviction concentration

The daily *timeline*.

A coordinated set of scheduled jobs runs each trading day, from morning health check through nightly content generation. Every job is managed by EventBridge with US Eastern timezone awareness.



Alert types.

ALERT	TRIGGER	AUDIENCE	CHANNEL
Daily digest	18:00 ET, trading days	All subscribers	Email + push
Sell alert	Trailing stop or regime exit	Position holders	Email + push
Warning alert	Within 3% of stop level	Position holders	Email + push

ALERT	TRIGGER	AUDIENCE	CHANNEL
Double signal	Timing + momentum cross within 3 days	All subscribers	Email
Intraday signal	Watchlist candidate clears threshold	Subscribers (not holding)	Email
Ticker health	Data issues detected at 07:00	Admin	Email

Email deliverability: SPF, DKIM, and DMARC authentication, RFC 8058 List-Unsubscribe headers for one-click unsubscription, and per-category preference controls.

Walk-forward *methodology*.

Walk-forward testing is the gold standard for trading-system validation. Unlike traditional backtesting, which optimizes across all available data and bakes in hindsight bias, walk-forward divides history into sequential periods and tests forward into data the optimizer has never seen.

Methodology.

- **Window.** A single continuous 21-year window (2007–2026), price returns.
- **Optimization.** Parameters are locked, then applied forward; no per-period retuning.
- **Trading.** Enters fresh ensemble signals as they fire; the universe is periodically re-ranked and existing positions are checked against the refreshed top tier.
- **Position carry.** Carry-on semantics: qualifying positions survive period boundaries rather than being mechanically liquidated.
- **Universe.** Top 100 symbols by 60-day volume per period; the production scanner runs the full universe daily.
- **Strategy.** The locked RigaCap Momentum Strategy timing-and-momentum ensemble (20 positions × 4.5%, inverse-volatility sizing, 30% trail, regime filter, Cascade Guard).
- **Data integrity.** Survivorship-free, point-in-time data from 2016 onward; the pre-2016 extension carries a survivorship caveat.

What we publish.

We publish one continuous window — 2007 through 2026 — with the crisis years and the bad years on the record, plus a fully held-out final 24 months as recent

confirmation. Every figure in the next section reflects one of those runs — we identify which one beside each figure.

Zero hindsight bias. At each step, the harness knows nothing about what happens next. Parameters are fixed in advance; the strategy then trades forward into unseen market conditions. The published figures are out-of-sample by construction.

What the walk-forward *shows*.

One continuous 21-year window (2007–2026), plus a fully held-out final 24 months. Both reported on clean indicator data, with the crisis years and the bad years on the record alongside the averages.

Headline numbers.

SERIES (2007–2026)	ANNUALISED	SHARPE	CALMAR	WORST DRAWDOWN
RigaCap Momentum Strategy	8.3%	0.73	0.43	19%
Raw 12-month momentum (net of costs)	13.2%	0.69	—	57%
<i>S&P 500 (price-only)</i>	<i>9.8%</i>	<i>0.54</i>	<i>—</i>	<i>55%</i>
Held-out last 24 months (Jun 2024–May 2026)	+32.0%	2.20	3.76	8.5%
<i>S&P 500, same 24-month window</i>	<i>+19.9%</i>	<i>1.18</i>	<i>1.05</i>	<i>19.0%</i>

Source: continuous 21-year walk-forward backtest, clean indicator data, price returns. Data from 2016 onward is survivorship-free and point-in-time; the pre-2016 extension carries a survivorship caveat. The live track record began June 11, 2026 with a fresh \$100k model book — all figures here are backtested.

Where the edge actually is.

On raw return, the strategy trails both the index and raw momentum over 21 years. The edge is risk: a 19% worst drawdown against 55% for the index and 57% for raw momentum, a 0.43 monthly correlation to the S&P, and an average month of –1.0%

when the index falls (the index averages -3.9%) versus $+1.7\%$ when it rises. The strategy is built to be held through a full cycle — including by investors who would panic-sell anything that loses half its value.

Cascade Guard contribution.

Cascade Guard is validated against a no-Cascade-Guard counterfactual run. A prior configuration showed a meaningful return-quality contribution; attribution for the current production configuration is being re-measured on the 21-year window before we publish a number. The drawdown effect is essentially neutral — we do not claim drawdown protection from Cascade Guard; the regime exit is what handles drawdown.

Crisis capital preservation

2008: -0.5% while the S&P 500 fell -37.7% (regime filter in cash). 2022: -7.5% against -19.9% . The clearest empirical evidence the regime-aware design earns its keep. The honest wart: 2018, a whipsaw year, came in at -12.3% against -7.0% .

Bull participation

2020: $+31.9\%$ against $+15.2\%$ for the S&P 500 — back in for the recovery after sidestepping the worst of the crash. The held-out last 24 months ran $+32.0\%$ annualised against $+19.9\%$.

How we trade.

Most entries are exited at the trailing stop, by design. The system does not need to be right most of the time. It needs winners to compound and losers to be cut on a fixed rule — and that is what the wide 30% trailing stop and exit priority deliver, spread across up to twenty inverse-volatility-sized positions.

Winners run, losers cut

The wide trailing stop lets profitable positions compound through normal volatility rather than capping them, while exits on the rule keep losses bounded.

Diversification, not concentration

Twenty smaller positions instead of a concentrated handful. The bar at entry remains high — the gauntlet rejects far more candidates than it admits — but no single name can sink the book.

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Source data: continuous 21-year walk-forward backtest (2007–2026, price returns) on clean indicator data, run June 2026. Data from 2016 onward is survivorship-free and point-in-time; pre-2016 data carries a survivorship caveat. The live track record began June 11, 2026 with a fresh \$100k model book. Production parameters are the locked RigaCap Momentum Strategy set: up to twenty positions sized at roughly 4.5% each with inverse-volatility weighting, a 30% trailing stop, the market-regime filter, Cascade Guard, and the timing-and-momentum ensemble described in this document.